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JOINT IMF/WORLD BANK DEBT SUSTAINABILITY ANALYSIS 2014

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Tonga remains at a moderate risk of debt distress. Despite some worsening of the underlying debt dynamics owing to the reconstruction needs from a recent cyclone, the risk rating continued to be "moderate." This was also affected by technical changes under the new DSA framework, namely higher discount factor and stronger focus on debt absorptive capacity of remittances.[1] The public sector DSA shows dynamics that are largely similar to those of public and publicly guaranteed external debt. Tonga's external debt sustainability continues to be vulnerable to external shocks. Among the different stress tests and alternative scenarios, a significant depreciation and slowdown in exports generate the highest debt ratios. This result underscores the importance of policies to promote export growth and diversification, and to hedge foreign exchange risks on external debt.

[1] The Country Policy and Institutional Assessment (CPIA) rating for Tonga remains at a medium level (Tonga's three year average CPIA score is 3.45).

BACKGROUND

1. Tonga is a small open economy with a narrow production base. It exports a few agricultural and fishery products, and imports food and services of numerous types. Its high reliance on remittances and foreign aid, which are subject to large fluctuations, poses challenges.

2. Economic growth is expected to pick up gradually (Box 1). In January 2014, a tropical cyclone hit Tonga. The resulting output loss will be more than offset by reconstruction. After recording between 0 and 1 percent in FY2012/13 (fiscal year ends on June 30), real GDP growth is expected to recover to 1½ percent in FY2013/14, and further to 3½ percent in FY2014/15, led by reconstruction from the cyclone. This recovery is also supported by rising agricultural production and an improving outlook for tourism. Over the longer-term, growth is expected to gradually converge to the historical trend of 1¾ percent, driven by remittances, tourism receipts, as well as credit growth. Risks to the near-term outlook are tilted to the downside.

3. Tonga's total debt stock (mainly external) rose from 30 percent of GDP in FY2007/08 to 42 percent of GDP in FY2012/13. It is expected to remain substantially above 30 percent in the medium term. The increase in debt levels, so far, has been mainly due to two large loans from a bilateral donor contracted in 2007 and 2010.[2] They account for about 60 percent of the outstanding external debt, while the long-term loans from international financial institutions (IFIs) account for only about 35 percent. The repayment terms for the 2007 loan to support the reconstruction of Nuku'alofa is currently under discussion by the relevant parties, and there are clear signs that the grace period will be extended, although a formal agreement has yet to be announced. The baseline scenario builds in an additional 5-year grace period.

4. Cyclone Ian-related reconstruction activity is expected to weaken the fiscal position. Assessment of the impact of the cyclone suggests property damage of about 10 percent of GDP and a widening of the overall fiscal balance by 1.5 percentage points of GDP to a deficit of 0.6 percent of GDP in FY2013/14. The effect of the cyclone will linger through FY2014/15, raising the overall deficit by about 1 percentage point compared to the no-cyclone forecast—to 0.7 percent of GDP. The projected fiscal cost relating to the cyclone, about 7 percent of GDP or 60 million pa'anga, is broadly met by funding from donor agencies.[3]

5. In the medium term, the overall fiscal balance is expected to return to a moderate surplus, starting in FY 2017/18. On current policies, this would lead to the gradual decline of the debt-to-GDP ratio to around 30–35 percent towards the end of the DSA timeframe (2034).

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[2] The loan for reconstruction of the central business district was signed in November 2007, and the loan for road improvement was signed in February 2010. These loans bear a 2 percent interest rate, 5-year grace period, and 20-year maturity.

[3] They consist of grants (about 70 percent), loans (about 20 percent), and disaster trust fund (about 10 percent). The new external loans are consistent with the authorities' "no new loan policy" as they are due to the World Bank financing of an equal grant-loan mix for countries with moderate risk of debt distress. The impact of these new external loans on debt sustainability is very small.

Box 1. Key Macroeconomic Assumptions

Real GDP growth is projected at 1½ percent in FY2013/14 and at 3½ percent in 2014/15, before reverting to the historical trend of 1.7 percent in the long term. The near-term pick up reflects the boosts provided by the reconstruction from the cyclone. The driver of the convergence of the underlying growth to the historical trend would be remittances, tourism receipts, and credit growth.

The overall fiscal balance is expected to be in deficit in the near term—given the 5 percent public sector salary increase in FY2013/14 and the extra fiscal costs related to the cyclone—but it will gradually move into small surplus toward the end of the medium term led by wage restraint and revenue reform. The continued efforts in improving tax policy and administration are expected to improve the tax-to-GDP ratio to 17.4 percent, and current expenditures are expected to decrease from about 27.2 percent of GDP in FY2013/14 to 22.5 percent over the medium term.[1] Capital expenditure, after elevating to about 5 percent of GDP in FY2014/15 will return to 2 percent over the medium term. Current and capital expenditures are assumed to be broadly stable as a percent of GDP in the long term, with the former about 21½ percent and the latter about 2 percent of GDP. The primary fiscal balance will average about 1 percent of GDP in the longer term.

Grants and projected loan disbursements are assumed to be mostly spent on development projects and associated maintenance. The external financing from the World Bank and the Asian Development Bank is expected to continue to be a 50 percent grant and 50 percent loan mix in the first eight years or so of the forecasting horizon. Domestic borrowing is assumed to be mainly led by gross financing requirement.

The current account deficit is projected to decline from 5½ percent of GDP in FY2012/13 to 3¼ percent in FY2013/14. This change is mainly due to higher remittances and official transfers that more than enough offset the higher imports. However, the current account deficit will increase to about 6 percent in FY2015/16—2016/17, mainly owing to the hosting of the Games. Afterward, it is expected to steadily decline to average of 3 percent in the long term.

Exports of goods are projected to remain stable at around 4 percent of GDP over the medium term. However, imports of goods are projected to reach 34 percent of GDP in the medium term and increase further to around 40 percent of GDP in the long term.

Remittances and tourism are projected to recover gradually, along with the global economic recovery. Remittances will increase from 15½ percent of GDP in FY2013/14 to about 19 percent of GDP over the medium term, and tourism receipts are projected to increase from 7 percent of GDP in FY2013/14 to about 10 percent over the medium term.

Tonga: Key Macroeconomic Assumptions (In percent)

	2013 DSA	2014 DSA
	2013-18 AVG	2014-19 AVG
Real GDP growth	1.5	2.3
GDP deflator in U.S. dollar terms (change)	1.5	1.5
Effective interest rate	1.9	1.5
Growth of exports of G&S (U.S. dollar terms)	7.6	5.6
Growth of imports of G&S (U.S. dollar terms)	2.9	5.9
Grant element of new public sector borrowing	31.3	54.1

Sources: Country authorities; and IMF staff estimates.

[1]Unwinding of the cyclone response cost and decrease in project grants account for about three quarter of this decline, and wage restraint the remainder.

6. Toward the outer years in the medium term, the South Pacific Games could pose risks if there are cost overruns or a shift toward debt-creating financing. The authorities estimate the cost of the Games at US$70 million which they hope to finance mainly by franchises and donations. The baseline scenario of the DSA uses the authorities' official costing and financing plan[4] which is fiscally prudent. However, historical experience suggests that cost overruns are not uncommon, and could create larger need for financing.

7. The updated DSA framework (DSF)[5] is applied to Tonga for the first time. The Country Policy and Institutional Assessment (CPIA) rating for Tonga remains at a medium level (Tonga's three year average CPIA score is 3.45). The new DSF puts greater emphasis on remittances as a source of foreign exchange and uses a higher discount rate (5 percent versus 3 percent in the previous framework), which contribute to technical improvements in debt and debt service indicators.[6]

EXTERNAL DSA

Baseline

8. Under the baseline scenario, the trajectories of external debt and debt service indicators remain comfortably below thresholds (Figure 1). The new DSA framework classifies Tonga as a large remittances case. In such a case, the external debt DSA is required to incorporate external debt absorption capacity generated by remittances into relevant debt burden and service capacity indicators, as well as distress thresholds tailored to them. With these adjustments, all relevant indicators shift down comfortably below their thresholds.

External Debt Indicators (In percent)

	Indicative Thresholds	Baseline Value in 2013/14
PV of PPG External Debt		
In percent of GDP	40	29
In percent of GDP and remittances	36	25
In percent of exports	150	148
In percent of exports and remittances	120	83
In percent of revenue	250	153
Debt Service		
In percent of exports	20	8
In percent of exports and remittances	16	4
In percent of revenue	20	8

Sources: Country authorities; and IMF staff estimates.

Alternative Scenario and Sensitivity Analysis

9. Stress tests show that the risk of debt distress is sensitive to a slowdown in exports or a large currency depreciation. For the debt services-to-revenue ratio (Figure 1), a onetime depreciation shock causes the indicator to breach the threshold slightly in FY2018/19, when the

[4] The authorities currently envisage to cover the operating cost (US$20 million) with the Games revenue, and the capital cost (US$50 million) with donations and franchises.

[5] See Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries and Revisiting the Debt Sustainability Framework for Low-Income Countries.

[6] Tonga's external DSA risk rating was reclassified from "high" to "moderate" last year mainly driven by an upgrade of CPIA rating.

extended grace period of the 2007 external loan provided by a bilateral donor (to support reconstruction of Nuku'alofa) is assumed to end. However, this breach is considered a borderline case. For all other indicators, there is no breach of thresholds even in the most extreme shock scenario.

10. To address the borderline issue, a probability approach was adopted. The new DSF contains a probability approach for assessing the risk of external debt distress in borderline cases, which is defined as one where the largest breach or near breach falls within a 10-percent band around the threshold.[7] Similar to the traditional approach, the risk of debt distress is sensitive to either a slowdown in exports or a significant depreciation, and the stress test results in multiple breaches of the thresholds. These results indicate a moderate risk of debt distress.

11. Coming on top of significant reconstruction needs, the Games could pose significant risks to external debt DSA. With nondebt financing covering the entire costs, the authorities' plan is fiscally prudent. If followed through, it would thus not affect Tonga's risk of debt distress—which is moderate. However, the experience of other countries has shown that actual cost often exceeds the original budget. As an illustration of risks, we simulated DSA assuming, hypothetically, a cost overrun financed by an external debt of US$170 million[8]. The simulation shows that Tonga will be at high risk of debt distress (see Appendix).

12. The historical scenario provides a substantially more sanguine picture than the baseline. This mainly reflects the higher growth of the GDP deflator in U.S. dollar terms applied to the historical scenario.[9] [10]

PUBLIC SECTOR DSA

Baseline

13. The public sector DSA shows largely similar dynamics to the external debt DSA (Figure 3). However, the overall picture is more benign reflecting relatively small levels (about 4 percent of nominal GDP) of public domestic debt. In the baseline scenario, the PV of the public debt-to-GDP ratio will remain slightly over 30 percent in the medium term, before it steadily declines throughout the projection period to about 20 percent. The debt services-to-revenue ratio shows a hump shape, similar to the case of the external debt DSA.

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[7] The probability approach incorporates a country's individual CPIA score and average GDP growth rate, whereas the traditional approach uses one of three discrete CPIA values (3.25 for weak performers, 3.50 for medium performers, and 3.75 for strong performers), and an average growth rate across LICs (for details see the Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries).

[8] The scenario assumes that the actual cost (US$190 million) would exceed the current official estimates (US$70 million) by about 3 times. The experience in other countries has shown that a large cost overrun of this tune is not improbable (see Box 2 of 2014 Tonga Staff Report for the 2014 Article IV Consultation).

[9] Four variables are set at their historical averages: (i) real GDP growth; (ii) growth of GDP deflator in U.S. dollar terms; (iii) non-interest current account in percent of GDP; and (IV) nondebt creating (FDI) flows.

[10] This means more accelerated growth in U.S. dollar nominal GDP given the same real growth rate.

Alternative Scenario and Sensitivity Analysis

14. Alternative scenarios underscore the importance of continued commitment to fiscal prudence. In contrast to the baseline scenario, the PV of public debt in the alternative scenarios—with the primary balance fixed at the FY2013/14 level, and with both real GDP growth and primary balance fixed at historical averages—fails to decline over the medium term. This demonstrates the need for consolidating the primary balance (from the current levels), as well as to resist reverting to the historical tendency of low growth and fiscal imbalance. Stress tests suggest that the public debt positions are particularly vulnerable to a significant depreciation. All indicators increase by a large margin with such a shock.

STAFF ASSESSMENT

15. Tonga remains at a moderate risk of debt distress. The underlying dynamics of the external debt has worsened somewhat compared to the previous DSA, due to the reconstruction needs of the recent cyclone. However, the risk rating continues to be moderate, partly owing to the technical changes under the new DSA framework, namely the higher discount factor and stronger focus on the debt absorptive capacity of remittances.

16. The public sector DSA shows broadly similar dynamics to public and publicly guaranteed external debt. However, the overall picture is more benign reflecting the relatively small levels of public domestic debt.

17. Tonga's external debt sustainability continues to be vulnerable to external shocks. Among the different stress tests and alternative scenarios, significant currency depreciation and a slowdown in exports generate the highest debt ratios. These results underscore the importance of policies to promote export growth and diversification and to hedge against exchange risks on external debt.

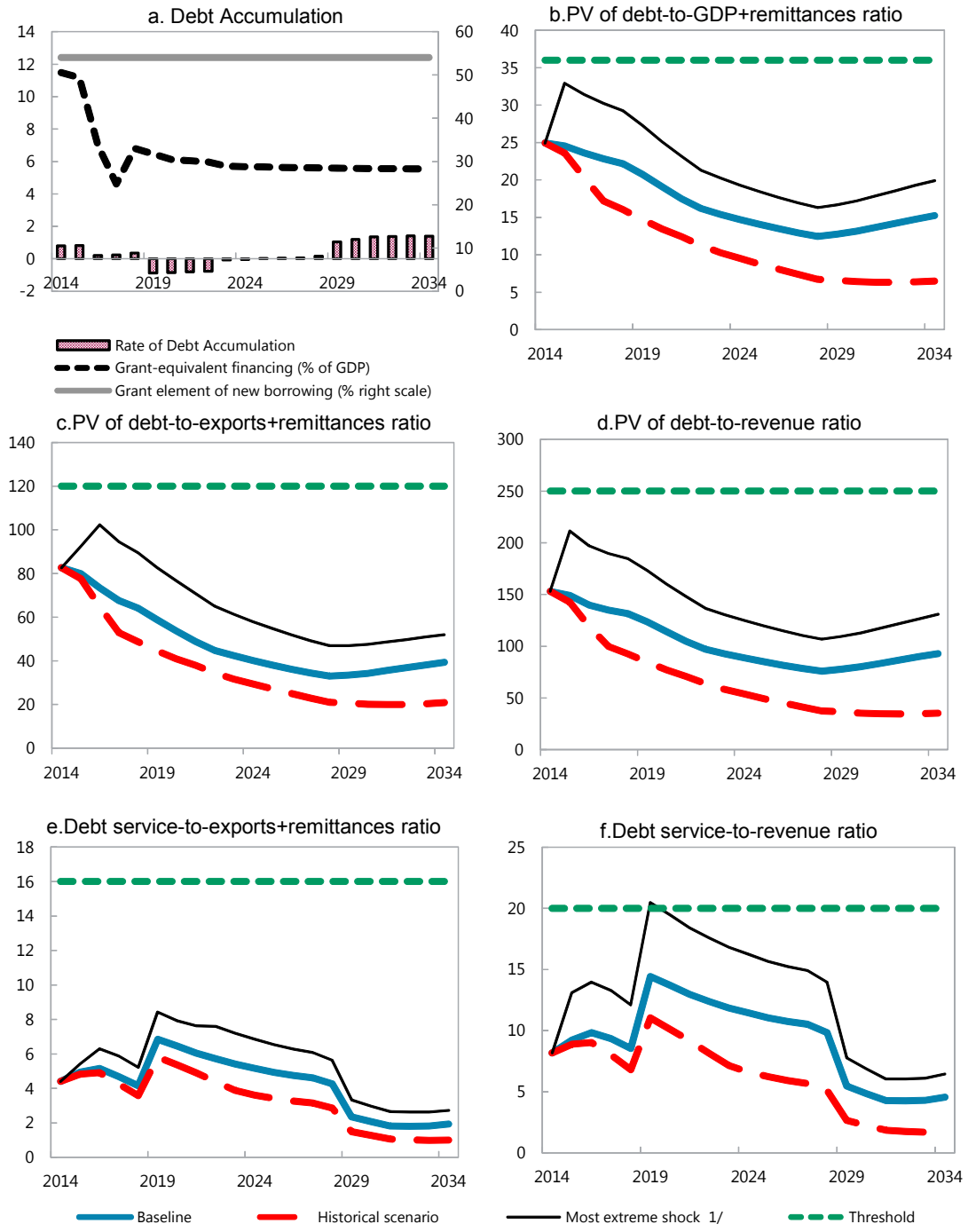
18. Well-designed policies to strengthen growth and improve fiscal management would help enhance debt sustainability. The authorities should continue to prioritize the use of fiscal space to high-impact growth-enhancing projects, while sustaining and expanding service delivery. Continued improvement of the public expenditure framework to attain higher efficiency is important in this regard. Developing a comprehensive debt management strategy, including using financial instruments to hedge exchange risks, is also a perennial priority.

19. Coming on top of significant reconstruction needs, the Games could pose significant risks to external debt DSA. As an illustration of risks, we simulated DSA assuming, hypothetically, a cost overrun financed by an external debt of US$ 170 million—which shows that Tonga will be at high risk of debt distress.

The Authorities' Views

20. The authorities agreed with the staff's assessment. They recognized that in addition to the existing debt burden, the cyclone damage potentially put more pressure on debt sustainability. They agreed with the need to build fiscal buffers and protect priority spending. The authorities intend to improve debt management framework.

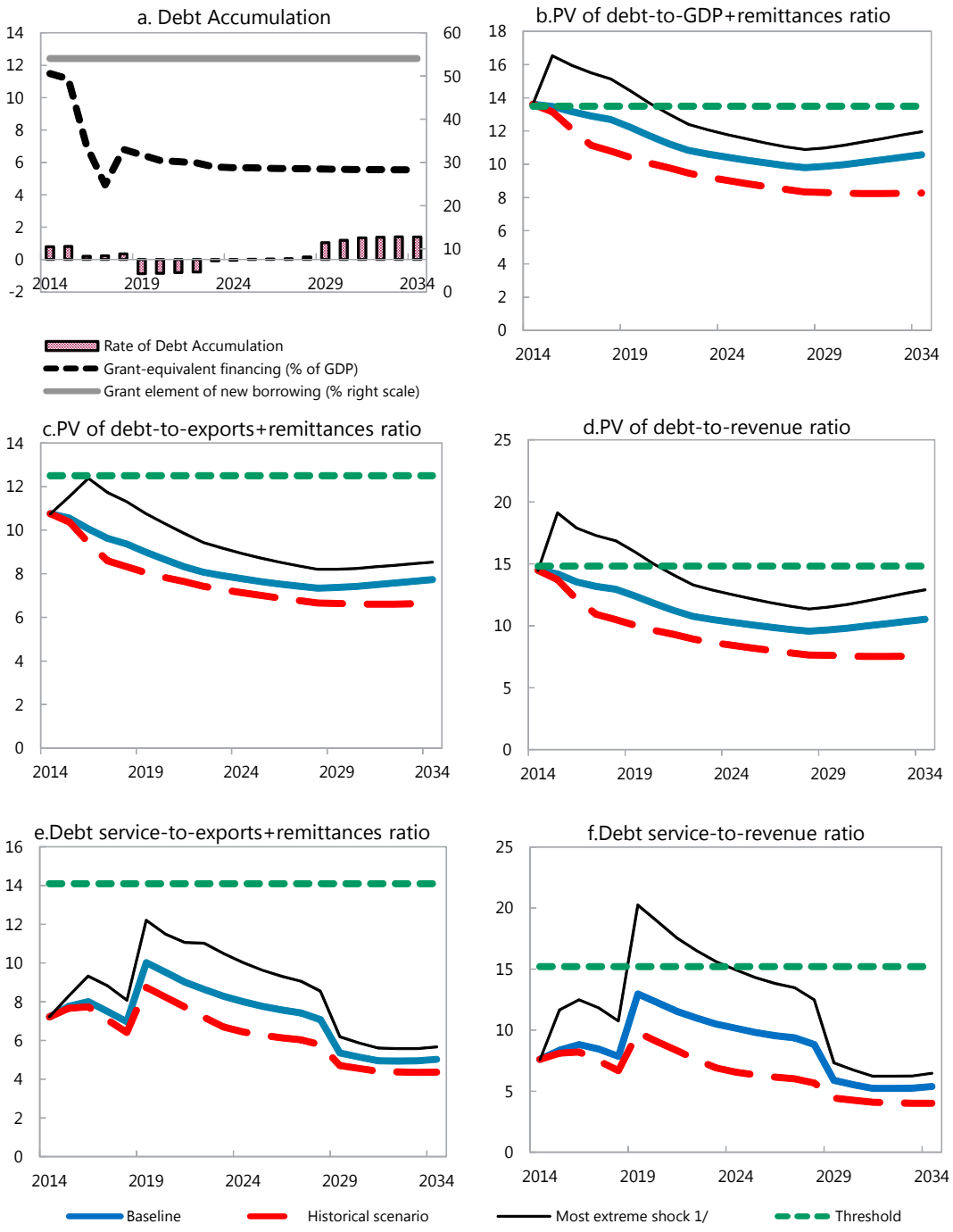
Figure 1. Tonga: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

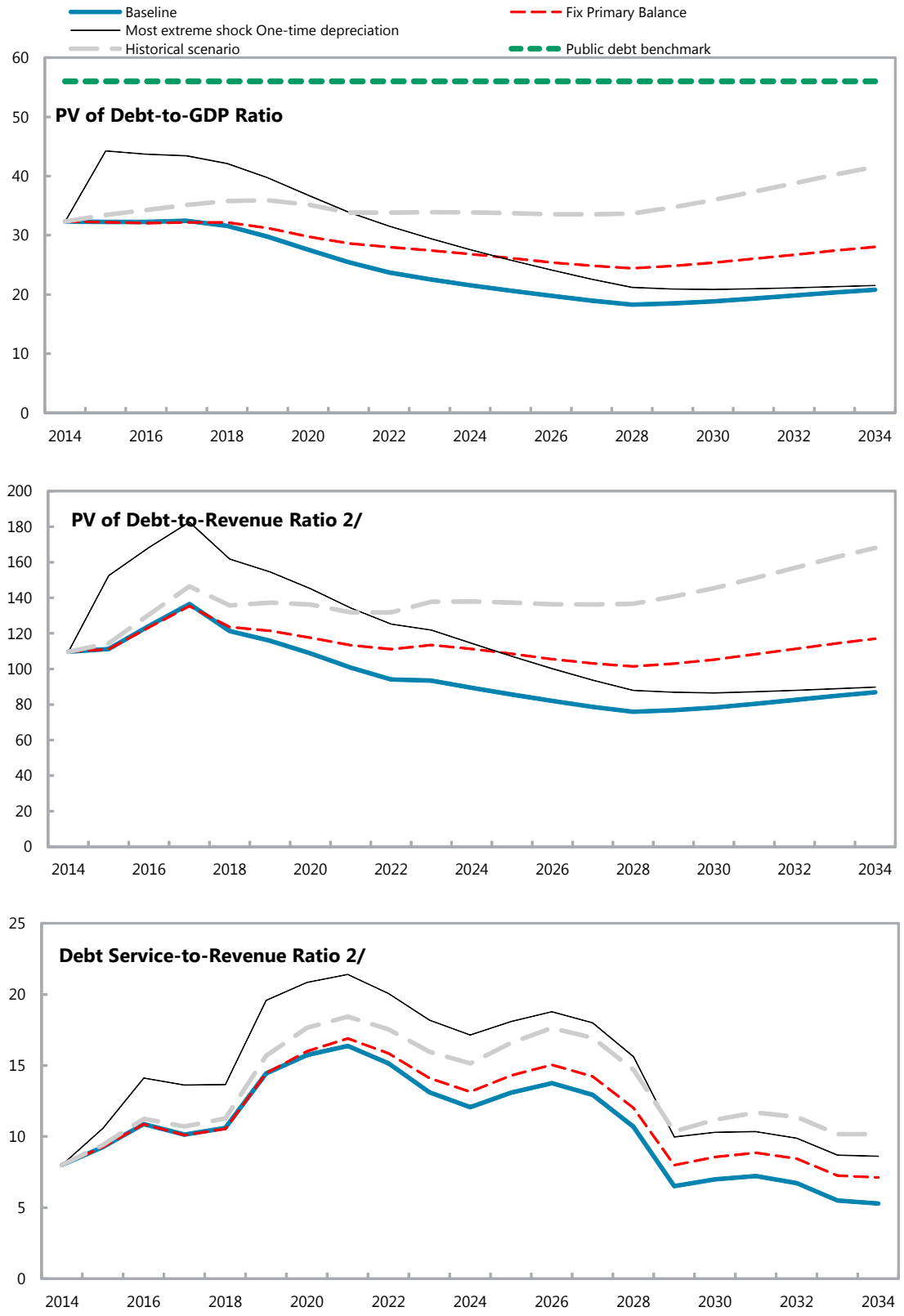
Figure 2. Tonga: Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 3. Tonga: Indicators of Public Debt Under Alternative Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 1a .Tonga: External Debt Sustainability Framework, Baseline Scenario, 2011-2034 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**36.0**	**41.9**	**41.4**			**41.0**	**40.3**	**39.0**	**37.9**	**37.0**	**35.2**		**28.6**	**35.4**	
of which: public and publicly guaranteed (PPG)	36.0	41.9	41.4			41.0	40.3	39.0	37.9	37.0	35.2		28.6	35.4	
Change in external debt	4.2	5.9	-0.5			-0.5	-0.6	-1.3	-1.1	-0.9	-1.8		-0.2	1.1	
Identified net debt-creating flows	-2.8	1.8	2.7			0.4	-0.7	2.6	3.2	-0.7	-2.7		-0.3	0.3	
Non-interest current account deficit	**2.2**	**6.1**	**4.6**	**4.2**	**3.5**	**2.7**	**2.4**	**5.4**	**5.8**	**1.8**	**0.3**		**2.5**	**2.8**	2.5
Deficit in balance of goods and services	32.7	36.3	25.5			29.0	31.3	31.7	30.7	30.0	29.0		30.4	32.7	
Exports	16.9	17.2	21.0			19.5	20.1	21.1	22.3	22.7	23.1		23.7	24.9	
Imports	49.5	53.5	46.5			48.5	51.3	52.8	52.9	52.7	52.1		54.0	57.6	
Net current transfers (negative = inflow)	-27.8	-27.4	-17.3	-30.1	5.3	-23.0	-25.5	-22.9	-21.5	-24.8	-25.3		-24.6	-26.7	-25.8
of which: official	-4.9	-3.3	-1.5			-6.4	-9.3	-5.6	-3.2	-5.4	-5.1		-3.7	-3.6	
Other current account flows (negative = net inflow)	-2.7	-2.9	-3.6			-3.4	-3.4	-3.4	-3.4	-3.4	-3.4		-3.2	-3.2	
Net FDI (negative = inflow)	**-1.5**	**-1.1**	**-3.4**	**-4.6**	**3.6**	**-2.4**	**-2.4**	**-2.4**	**-2.4**	**-2.5**	**-2.5**		**-2.6**	**-2.2**	-2.5
Endogenous debt dynamics 2/	**-3.5**	**-3.2**	**1.6**			**0.1**	**-0.7**	**-0.4**	**-0.1**	**-0.1**	**-0.5**		**-0.2**	**-0.3**	
Contribution from nominal interest rate	0.6	0.5	0.9			0.6	0.6	0.6	0.6	0.5	0.5		0.3	0.3	
Contribution from real GDP growth	-0.8	-0.2	-0.1			-0.6	-1.3	-1.0	-0.7	-0.6	-1.0		-0.5	-0.6	
Contribution from price and exchange rate changes	-3.2	-3.5	0.7			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**7.0**	**4.1**	**-3.2**			**-0.8**	**0.1**	**-3.9**	**-4.4**	**-0.2**	**0.9**		**0.1**	**0.8**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	28.9			28.8	28.3	27.5	26.8	26.2	24.7		17.7	18.7	
In percent of exports	...	...	137.6			147.9	141.1	130.5	120.3	115.4	106.9		74.9	74.9	
PV of PPG external debt	**...**	**...**	**28.9**			**28.8**	**28.3**	**27.5**	**26.8**	**26.2**	**24.7**		**17.7**	**18.7**	
In percent of exports	**...**	**...**	**137.6**			**147.9**	**141.1**	**130.5**	**120.3**	**115.4**	**106.9**		**74.9**	**74.9**	
In percent of government revenues	**...**	**...**	**154.6**			**153.2**	**149.4**	**139.9**	**134.8**	**131.5**	**123.6**		**88.9**	**93.0**	
Debt service-to-exports ratio (in percent)	**7.1**	**7.7**	**8.6**			**7.9**	**8.7**	**9.2**	**8.3**	**7.5**	**12.5**		**9.6**	**3.7**	
PPG debt service-to-exports ratio (in percent)	**7.1**	**7.7**	**8.6**			**7.9**	**8.7**	**9.2**	**8.3**	**7.5**	**12.5**		**9.6**	**3.7**	
PPG debt service-to-revenue ratio (in percent)	**6.2**	**7.3**	**9.6**			**8.2**	**9.2**	**9.8**	**9.3**	**8.5**	**14.4**		**11.4**	**4.5**	
Total gross financing need (Millions of U.S. dollars)	8.0	29.9	13.7			8.8	8.7	26.1	28.2	6.0	3.6		15.4	16.7	
Non-interest current account deficit that stabilizes debt ratio	-2.0	0.1	5.1			3.1	3.0	6.7	6.9	2.8	2.1		2.7	1.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.9	0.7	0.3	0.9	2.4	1.4	3.4	2.5	1.7	1.7	2.8	2.3	1.7	1.7	1.6
GDP deflator in US dollar terms (change in percent)	11.4	10.7	-1.8	8.0	8.1	2.4	1.5	1.6	1.8	2.0	0.0	1.5	2.5	2.5	2.6
Effective interest rate (percent) 5/	2.0	1.5	2.2	1.6	0.7	1.6	1.6	1.5	1.5	1.5	1.4	1.5	1.2	0.9	1.0
Growth of exports of G&S (US dollar terms, in percent)	51.9	13.6	20.2	10.0	19.0	-3.8	8.2	9.4	9.5	5.8	4.4	5.6	4.8	4.2	4.8
Growth of imports of G&S (US dollar terms, in percent)	17.8	20.5	-14.5	8.3	13.3	8.2	11.0	7.2	3.9	3.3	1.5	5.9	4.9	4.1	5.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	54.1	54.1	54.1	54.1	54.1	54.1	54.1	54.1	54.1	54.1
Government revenues (excluding grants, in percent of GDP)	19.5	18.1	18.7			18.8	19.0	19.6	19.9	19.9	20.0		20.0	20.1	20.0
Aid flows (in Millions of US dollars) 7/	28.8	44.2	25.2			58.7	61.3	40.1	28.5	42.0	41.3		50.0	75.3	
of which: Grants	28.8	44.2	25.2			51.6	50.8	33.3	21.3	34.5	33.4		29.8	42.5	
of which: Concessional loans	0.0	0.0	0.0			7.1	10.6	6.8	7.2	7.5	7.9		20.2	32.9	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			11.5	11.2	7.0	4.6	6.8	6.5		5.7	5.5	5.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			94.5	92.1	92.2	88.4	91.8	91.2		81.5	80.0	82.7
Memorandum items:															
Nominal GDP (Millions of US dollars)	423.0	471.6	464.5			482.2	505.8	527.0	545.9	566.3	582.1		716.8	1087.3	
Nominal dollar GDP growth	14.6	11.5	-1.5			3.8	4.9	4.2	3.6	3.7	2.8	3.8	4.3	4.3	4.3
PV of PPG external debt (in Millions of US dollars)			135.2			138.8	142.7	143.7	144.9	146.8	141.8		126.1	201.3	
(PVt-PVt-1)/GDPt-1 (in percent)						0.8	0.8	0.2	0.2	0.3	-0.9	0.2	0.0	1.4	0.4
Gross workers' remittances (Millions of US dollars)	87.5	66.8	68.8			74.2	77.6	86.3	94.6	103.0	110.0		147.2	244.3	
PV of PPG external debt (in percent of GDP + remittances)	...	...	25.2			24.9	24.5	23.6	22.8	22.2	20.7		14.7	15.2	
PV of PPG external debt (in percent of exports + remittances)	...	...	80.7			82.6	80.0	73.4	67.6	64.1	58.8		40.1	39.4	
Debt service of PPG external debt (in percent of exports + remittances	...	...	5.0			4.4	4.9	5.2	4.7	4.2	6.9		5.2	1.9	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Tonga: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034
(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	29	28	27	27	26	25	**18**	19
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	29	27	23	20	18	17	**11**	7
A2. New public sector loans on less favorable terms in 2014-2034 2	29	29	28	28	28	26	**22**	31
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	29	30	30	29	28	27	**19**	20
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	29	30	32	31	31	29	**22**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	29	29	28	27	27	25	**18**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	29	29	28	27	27	25	**18**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	29	28	27	26	25	24	**17**	18
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	29	40	39	38	37	35	**25**	26
PV of debt-to-exports ratio								
Baseline	148	141	131	120	115	107	**75**	75
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	148	135	111	89	81	73	**45**	28
A2. New public sector loans on less favorable terms in 2014-2034 2	148	144	134	125	122	115	**94**	123
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	148	141	129	119	114	106	**74**	74
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	148	176	216	200	192	180	**130**	119
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	148	141	129	119	114	106	**74**	74
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	148	146	133	123	118	109	**77**	76
B5. Combination of B1-B4 using one-half standard deviation shocks	148	150	147	135	129	119	**83**	85
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	148	141	129	119	114	106	**74**	74
PV of debt-to-revenue ratio								
Baseline	153	149	140	135	132	124	**89**	93
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	153	143	119	100	93	85	**54**	35
A2. New public sector loans on less favorable terms in 2014-2034 2	153	152	144	140	139	133	**112**	153
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	153	156	152	146	142	133	**96**	101
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	153	157	162	157	153	145	**108**	103
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	153	151	143	138	134	126	**91**	95
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	153	154	143	138	134	126	**91**	94
B5. Combination of B1-B4 using one-half standard deviation shocks	153	149	137	131	128	119	**86**	92
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	153	211	197	190	185	173	**125**	131

Table 1b.Tonga: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	8	9	9	8	7	12	**10**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	8	8	8	7	6	10	**6**	1
A2. New public sector loans on less favorable terms in 2014-2034 2	8	9	9	9	8	13	**11**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	8	9	9	8	7	12	**10**	4
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	8	10	13	12	11	18	**15**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	8	9	9	8	7	12	**10**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	8	9	9	8	8	13	**10**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	8	9	11	10	9	15	**11**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	8	9	9	8	7	12	**10**	4
Debt service-to-revenue ratio								
Baseline	8	9	10	9	9	14	**11**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	8	9	9	8	7	11	**7**	2
A2. New public sector loans on less favorable terms in 2014-2034 2	8	9	10	10	9	15	**13**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	8	10	11	10	9	16	**12**	5
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	8	9	10	10	9	15	**13**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	8	9	10	10	9	15	**12**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	8	9	10	9	9	14	**12**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	8	9	10	9	9	15	**11**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	8	13	14	13	12	20	**16**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	52	52	52	52	52	52	**52**	52

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Tonga: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011-2034

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	41.3	46.5	45.1			44.5	44.3	43.8	43.6	42.3	40.3		32.4	37.6	
of which: foreign-currency denominated	36.0	41.9	41.4			41.0	40.3	39.0	37.9	37.0	35.2		28.6	35.4	
Change in public sector debt	1.3	5.2	-1.4			-0.6	-0.2	-0.5	-0.2	-1.3	-2.1		-0.4	1.0	
Identified debt-creating flows	0.3	2.3	-1.0			-0.8	-1.2	-0.7	-0.8	-2.6	-2.3		-2.0	-1.8	
Primary deficit	6.6	2.0	-0.1	0.7	3.0	-0.2	-0.1	0.2	-0.1	-1.8	-1.8	-0.6	-1.1	-0.7	-1.1
Revenue and grants	26.3	27.4	24.1			29.5	29.0	26.0	23.8	26.0	25.7		24.1	24.0	
of which: grants	6.8	9.4	5.4			10.7	10.0	6.3	3.9	6.1	5.7		4.2	3.9	
Primary (noninterest) expenditure	32.9	29.5	24.0			29.3	28.9	26.1	23.7	24.2	23.8		23.0	23.3	
Automatic debt dynamics	-6.4	0.8	0.1			-0.6	-1.1	-0.8	-0.8	-0.8	-0.4		-0.9	-1.1	
Contribution from interest rate/growth differential	-1.2	-0.3	0.3			-0.5	-1.4	-1.1	-0.8	-0.9	-1.2		-0.7	-1.0	
of which: contribution from average real interest rate	0.0	0.0	0.4			0.1	0.1	0.0	-0.1	-0.1	0.0		-0.2	-0.3	
of which: contribution from real GDP growth	-1.1	-0.3	-0.1			-0.6	-1.5	-1.1	-0.7	-0.7	-1.2		-0.6	-0.6	
Contribution from real exchange rate depreciation	-5.2	1.1	-0.3			-0.1	0.2	0.3	0.1	0.1	0.8		...	...	
Other identified debt-creating flows	0.1	-0.6	-0.9			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.1	-0.6	-0.9			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.0	2.9	-0.4			0.2	1.0	0.2	0.6	1.3	0.2		1.6	2.8	
Other Sustainability Indicators															
PV of public sector debt	...	...	32.6			32.3	32.3	32.3	32.5	31.6	29.8		21.6	20.8	
of which: foreign-currency denominated	...	...	28.9			28.8	28.3	27.5	26.8	26.2	24.7		17.7	18.7	
of which: external	...	...	28.9			28.8	28.3	27.5	26.8	26.2	24.7		17.7	18.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	9.4	3.9	2.8			2.2	2.6	3.0	2.4	0.9	1.9		1.8	0.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	135.0			109.7	111.2	124.3	136.5	121.4	115.9		89.5	86.9	
PV of public sector debt-to-revenue ratio (in percent)	...	...	174.2			172.2	170.2	164.2	163.4	158.5	149.2		108.1	103.8	
of which: external 3/	...	...	154.6			153.2	149.4	139.9	134.8	131.5	123.6		88.9	93.0	
Debt service-to-revenue and grants ratio (in percent) 4/	10.7	7.0	12.2			8.0	9.3	10.9	10.1	10.6	14.4		12.1	5.3	
Debt service-to-revenue ratio (in percent) 4/	14.4	10.6	15.8			12.6	14.2	14.4	12.1	13.9	18.6		14.6	6.3	
Primary deficit that stabilizes the debt-to-GDP ratio	5.2	-3.2	1.3			0.4	0.2	0.6	0.2	-0.6	0.2		-0.7	-1.7	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.9	0.7	0.3	0.9	2.4	1.4	3.4	2.5	1.7	1.7	2.8	2.3	1.7	1.7	1.6
Average nominal interest rate on forex debt (in percent)	2.0	1.5	2.2	1.6	0.7	1.6	1.6	1.5	1.5	1.5	1.4	1.5	1.2	0.9	1.0
Average real interest rate on domestic debt (in percent)	-2.0	1.6	4.3	-2.0	4.9	1.3	3.6	2.6	1.8	1.4	3.0	2.3	1.3	1.3	1.2
Real exchange rate depreciation (in percent, + indicates depreciation	-16.8	3.2	-0.6	-4.8	8.3	-0.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	5.8	2.4	0.7	5.3	5.2	1.5	1.7	2.8	3.6	4.0	2.3	2.7	4.1	4.1	4.2
Growth of real primary spending (deflated by GDP deflator, in percer	0.1	-0.1	-0.2	0.0	0.2	0.2	0.0	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	54.1	54.1	54.1	54.1	54.1	54.1	54.1	54.1	54.1	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Tonga: Sensitivity Analysis for Key Indicators of Public Debt 2014-2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	32	32	32	32	32	30	22	21
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	32	33	34	35	36	36	34	42
A2. Primary balance is unchanged from 2014	32	32	32	32	32	31	27	28
A3. Permanently lower GDP growth 1/	32	32	33	33	33	31	26	34
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	32	34	37	38	38	37	32	38
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	32	34	36	36	35	33	25	23
B3. Combination of B1-B2 using one half standard deviation shocks	32	35	37	38	37	36	30	34
B4. One-time 30 percent real depreciation in 2015	32	44	44	43	42	40	28	22
B5. 10 percent of GDP increase in other debt-creating flows in 2015	32	37	37	38	37	35	26	24
PV of Debt-to-Revenue Ratio 2/								
Baseline	110	111	124	137	121	116	89	87
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	110	114	131	146	136	137	138	168
A2. Primary balance is unchanged from 2014	110	111	123	135	124	122	111	117
A3. Permanently lower GDP growth 1/	110	112	126	140	125	121	105	138
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	110	116	138	157	142	140	131	158
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	110	118	139	152	135	130	103	98
B3. Combination of B1-B2 using one half standard deviation shocks	110	118	139	156	141	138	124	142
B4. One-time 30 percent real depreciation in 2015	110	153	168	183	162	155	114	90
B5. 10 percent of GDP increase in other debt-creating flows in 2015	110	129	144	158	141	135	108	102
Debt Service-to-Revenue Ratio 2/								
Baseline	8	9	11	10	11	14	12	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	9	11	11	11	16	15	10
A2. Primary balance is unchanged from 2014	8	9	11	10	11	14	13	7
A3. Permanently lower GDP growth 1/	8	9	11	10	11	15	13	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	8	10	12	11	12	16	15	9
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	8	9	11	11	12	15	13	6
B3. Combination of B1-B2 using one half standard deviation shocks	8	9	12	11	12	16	14	9
B4. One-time 30 percent real depreciation in 2015	8	11	14	14	14	20	17	9
B5. 10 percent of GDP increase in other debt-creating flows in 2015	8	9	11	12	12	16	13	6

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

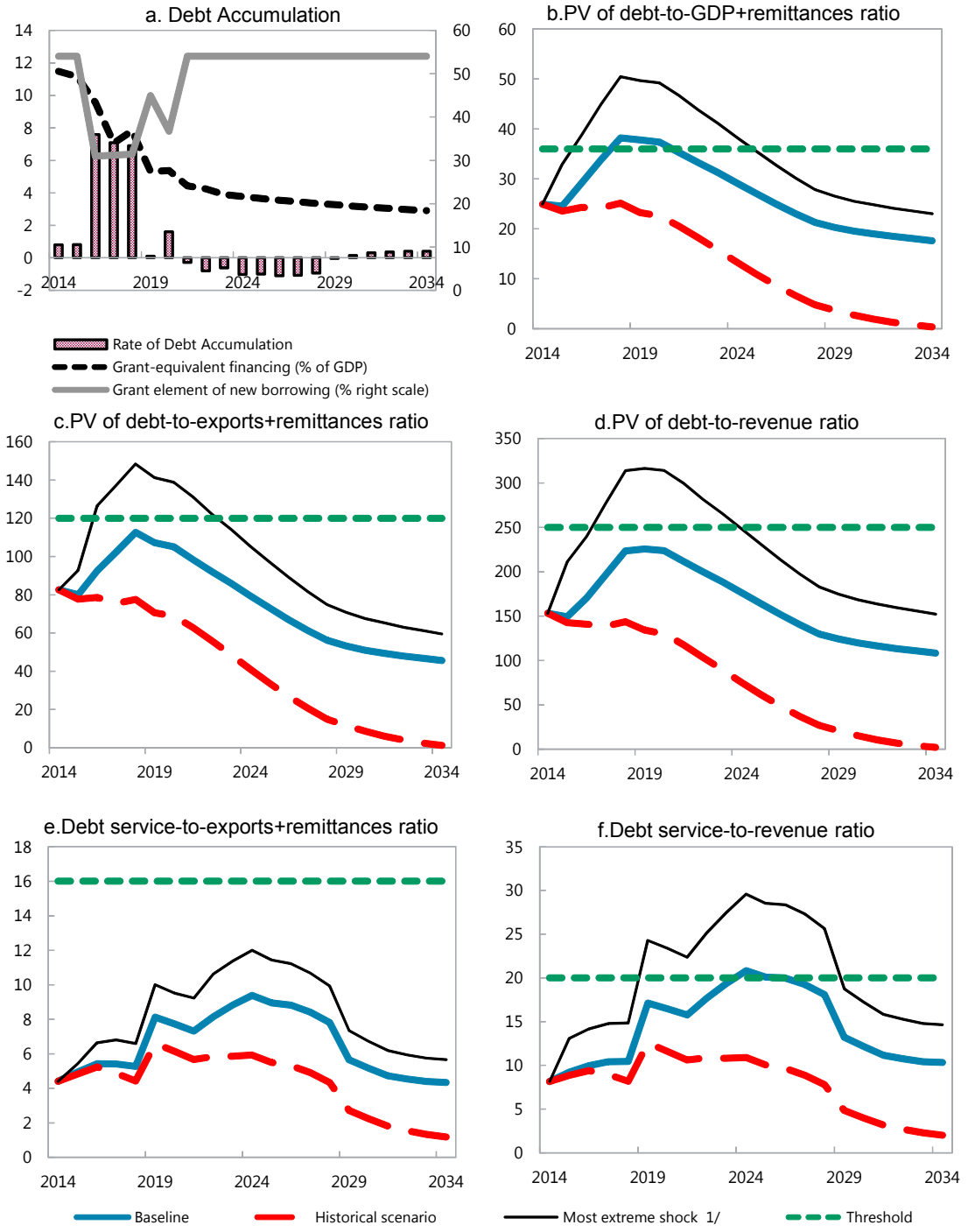
Appendix. The 2019 Pacific Games: High-Cost Scenario

Tonga will host the 2019 Pacific Games. While the authorities' original plan is fiscally prudent, international experience suggests that cost overrun often takes place. As an illustration of risks, we simulate a DSA assuming, hypothetically, a cost overrun (by US$120 million) financed by an external debt (US$170 million). We assume the hypothetical external loans will be made on terms and conditions similar to the loans provided by China's EXIM bank in recent years (2 percent interest rate, 20-year maturity, and 5-year grace period).

The macroeconomic assumptions remain broadly similar, compared to the baseline. In the run-up to the Games, the projects of building new sports facilities and improving existing accommodations and infrastructure would directly boost GDP. However, the growth rate decreases rapidly afterward, reflecting cooling-down effect after the pre-Games construction. During the game year of 2019, the positive effects brought by additional tourists and services provided will help spur a second round of growth windfall followed by a year of unwinding.

The simulated DSA causes Tonga's rating to change from moderate to high risk of debt distress. The present value of public and publicly guaranteed debt to GDP plus remittances ratio would breach its threshold for 3-4 years under the baseline (Figure A1. High-cost). Under stress tests, four out of five debt and debt service indicators would breach the relevant thresholds. In addition, the PV of public debt to GDP also breaches the threshold under a stress scenario.

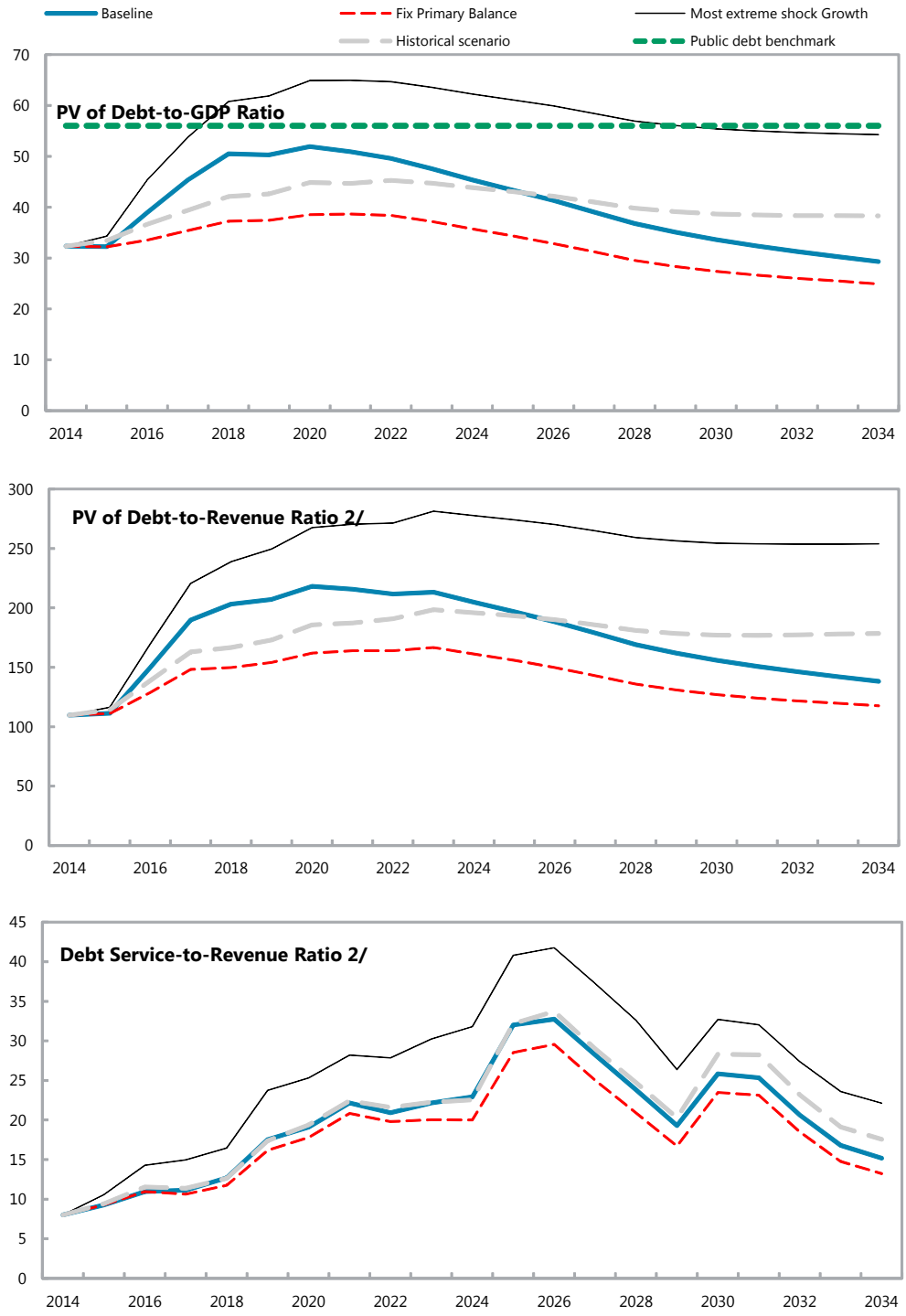
Figure A1. High-cost. Tonga: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure A2. High-cost. Tonga: Indicators of Public Debt Under Alternative Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.